

Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-09-13 07:30:01

Situation Summary

Stage IV: 6 of 30
Biomarker Count: 23 entries
Top Targets: ctDNA (7) · MSI/MMR (5) · NGS (3)
Breakthrough: 12 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED AUG 24, 2026

9

Liquid biopsy in colorectal cancer: redefining surgical decision-making

Liquid biopsy (LB) provides a minimally invasive method for detecting circulating tumor DNA (ctDNA) in colorectal cancer, offering insights into tumor biology that enhance postoperative management. This scoping review highlights LB's potential to inform surgical decision-making and improve risk stratification beyond traditional anatomical assessments.

pubmed.ncbi.nlm.nih.gov

HIGH

Stage III

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED AUG 18, 2026

7

Circulating tumor DNA in colorectal cancer: assay selection, clinical applications, and practical integration for gastroenterologists

Circulating tumor DNA (ctDNA) testing in colorectal cancer requires careful assay selection based on clinical intent, with tumor-informed platforms recommended for minimal residual disease (MRD) assessment and tumor-agnostic platforms for cases lacking tissue. Results should be interpreted considering biological and technical contexts, including platform sensitivity and quantitative variant allele frequency (VAF) trends. These principles are essential for effective integration of ctDNA testing in postoperative and surveillance settings.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED SEP 04, 2026

6

The Role of Biomarkers in Personalized Treatment of Gastrointestinal Cancers

Biomarkers such as KRAS, HER2, PD-L1, microsatellite instability (MSI), and circulating tumor DNA are essential for advancing personalized treatment strategies in gastrointestinal cancers, including colorectal cancer. This narrative review highlights their roles in diagnosis, prognosis, and treatment prediction, addressing the challenges posed by the biological complexity and late-stage diagnosis of these malignancies.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED AUG 21, 2026

6

Beyond tissue biopsy: ctDNA methylation analysis for prediction of colorectal cancer prognosis

ctDNA methylation analysis offers a novel approach for prognostic stratification in colorectal cancer, addressing limitations of mutation-based methods that focus solely on genetic alterations. This study highlights the potential of ctDNA methylation to provide deeper insights into tumor biology and improve postoperative risk assessment by better capturing molecular residual disease (MRD).

pubmed.ncbi.nlm.nih.gov

HIGH

Early Detection

MSI/MMR

VERIFIED

GUIDELINE

PUBLISHED SEP 10, 2026

5

Uptake of germline testing for Lynch Syndrome in patients with deficient mismatch repair/microsatellite-high colorectal cancer in the public hospital system in South Australia

In a retrospective review of 1,775 colorectal cancer patients in South Australia, the study assessed the uptake of germline testing for Lynch syndrome in those with deficient mismatch repair or microsatellite-high tumors. Despite guidelines recommending universal MMR/MSI screening, the study found variable rates of screening, germline testing, and genetics referrals, highlighting gaps in adherence to testing protocols for Lynch syndrome in clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED AUG 19, 2026

5

Early-Onset Colorectal Cancer With Liver-Only Metastases: A Retrospective Cohort Study Integrating Prospectively Collected Real-World Clinical and Molecular Data From an Australian National Database (2009-2024) to Guide Treatment Planning

Early-onset colorectal cancer (EOCRC) exhibits distinct biological differences compared to liver-only metastatic colorectal cancer (LOCRC), including variations in sex distribution and BRAF mutation rates. In a retrospective cohort study utilizing real-world clinical and molecular data, liver resection was linked to improved survival in LOCRC. Treatment benefits varied based on age, primary tumor site, and the timing of liver metastases, indicating the need for tailored treatment planning.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

EGFR

VERIFIED

RESEARCH

PUBLISHED SEP 07, 2026

4

Second-line raltitrexed plus irinotecan (SALIRI) with bevacizumab, cetuximab, or alone for metastatic colorectal cancer: a prospective multicenter cohort-based secondary analysis of the SALLY trial

SALIRI, a combination of raltitrexed and irinotecan, demonstrated feasibility and tolerability as a second-line therapy for metastatic colorectal cancer when combined with bevacizumab or cetuximab in a nonrandomized secondary analysis of the SALLY trial. Favorable outcomes were noted in the SALIRI + cetuximab subgroup, though results should be interpreted with caution due to nonrandom treatment allocation. Further prospective, biomarker-driven randomized studies are needed to clarify the optimal use of SALIRI-based combinations.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

NGS

VERIFIED

GENOMIC PROFILING

PUBLISHED AUG 26, 2026

4

Integrated NGS and ddPCR analysis of RAS/RAF mutations in metastatic colorectal cancer among Kazakhstani patients: a pilot study

KRAS mutations are the predominant genetic alterations in metastatic colorectal cancer, with BRAF mutations being less frequent. The study utilized targeted next-generation sequencing and droplet digital PCR to enhance mutation detection accuracy in FFPE samples. These findings elucidate the molecular characteristics of metastatic colorectal cancer in Kazakhstan and support the advancement of precision oncology strategies in clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Early Detection

MSI/MMR

VERIFIED

EARLY DETECTION

PUBLISHED SEP 11, 2026

2

A Lynch Syndrome Family With Germline MLH1 c.931A>G Showing Preserved Tumor MMR Immunostaining but MSI-H Status: A Case Report

A family with the germline MLH1 c.931A>G variant exhibited retained mismatch repair protein expression via immunohistochemistry despite showing microsatellite instability-high (MSI-H) status on tumor sequencing. This

discordance highlights potential complexities in Lynch syndrome screening and underscores the need for comprehensive testing approaches in colorectal cancer cases.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

GENOMIC PROFILING

PUBLISHED SEP 10, 2026

⚡ 2

Reference-Free Microsatellite Instability Detection from Tumor Sequencing Using Intrasample Variability Modeling

PROMIS is a tumor-only, reference-free pipeline for detecting microsatellite instability (MSI) using a discrete mixture model to analyze intrasample repeat-length distributions. This method addresses limitations of existing MSI detection techniques that require matched normal samples or reference panels, enhancing assay portability across different sequencing centers. The study demonstrates the feasibility of using PROMIS for MSI detection in colorectal cancer, potentially improving biomarker testing accessibility.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Early Detection

KRAS

VERIFIED

GENOMIC PROFILING

PUBLISHED SEP 09, 2026

⚡ 2

Fulminant development of a giant pelvic metastasis from microsatellite-stable early-onset sigmoid colon cancer with KRAS and TP53 co-mutations: a case report

Early-onset microsatellite-stable colorectal cancer with concurrent KRAS and TP53 mutations can demonstrate hyper-progressive behavior, leading to significant pelvic metastasis. Negative screening results do not exclude interval colorectal cancer. Clinicians should remain vigilant for primary gastrointestinal malignancies in female patients presenting with large pelvic masses. Timely multi-gene next-generation sequencing (NGS) panel testing is essential for guiding treatment decisions.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

KRAS

VERIFIED

RESEARCH

PUBLISHED SEP 03, 2026

⚡ 2

Unlocking HER2 as a target in KRAS-mutated rectal cancer: a case of significant response to zanidatamab-based regimen

HER2 inhibition may be an effective therapeutic strategy in KRAS-mutated colorectal cancer, as demonstrated in a case involving a significant response to a zanidatamab-based regimen. This finding supports the potential of zanidatamab in conversion therapy for patients with historically refractory disease.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage II

KRAS

VERIFIED

RESEARCH

PUBLISHED AUG 27, 2026

⚡ 2

KRAS Mutation Detection by Real-Time and Digital PCR in Tumor Tissue and Plasma and Its Association with Survival in Stage II-IV Colorectal Cancer: A Kazakhstan Cohort Study

KRAS mutation detection in tumor tissue and plasma was assessed using real-time and digital PCR in a Kazakhstan cohort of Stage II-IV colorectal cancer. The study found that variant-level reporting provides more prognostic information than binary calls. Additionally, a higher detection rate in plasma using digital PCR was observed, suggesting the need for further prospective studies with standardized mutation panels.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

EGFR

VERIFIED

RESEARCH

PUBLISHED SEP 12, 2026

⚡ 1

Quantitative dot blot method for measuring EGFR protein levels in patients with colorectal cancer

A quantitative dot blot (QDB) assay was developed to measure EGFR protein levels in colorectal cancer specimens. This method provides an objective assessment of EGFR as a predictive biomarker, facilitating both retrospective and prospective evaluations of anti-EGFR therapies in clinical trials.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

GENOMIC PROFILING

PUBLISHED SEP 03, 2026

⚡ 1

Stromal ARHGEF15 Correlates With Inflammatory EMT and Stromal-Immune Crosstalk During Inflammatory Bowel Disease-To-Colorectal Cancer Progression

Stromal ARHGEF15 expression correlates with inflammatory epithelial-mesenchymal transition (EMT) and stromal-immune interactions in the progression from inflammatory bowel disease (IBD) to colorectal cancer (CRC). This study utilized retrospective analysis of single-cell transcriptomic datasets from intestinal biopsies, highlighting the role of ARHGEF15 in the malignant transformation associated with chronic intestinal inflammation. Understanding this correlation may inform therapeutic strategies targeting the inflammatory pathways in IBD-related CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

NGS

VERIFIED

GENOMIC PROFILING

PUBLISHED SEP 02, 2026

⚡ 1

Lynch syndrome-associated urothelial carcinoma: clinical and molecular findings from a single-institution cohort

Lynch syndrome-associated urothelial carcinoma (LS-UC) presents unique clinical and molecular characteristics. A retrospective single-center study assessed patients with Lynch syndrome and histologically confirmed urothelial carcinoma, focusing on FGFR3 alterations. The findings clarify the prevalence and clinical implications of FGFR3 mutations in LS-UC, contributing to the understanding of this rare entity.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

GENOMIC PROFILING

PUBLISHED AUG 22, 2026

⚡ 1

Prevalence and clinical impact of PIK3CA and PTEN alterations from reflex genomic testing of newly diagnosed colorectal cancer: a population-based series

In a population-based series, the prevalence of PIK3CA and PTEN alterations in newly diagnosed colorectal cancer was found to be 14.6%, lower than the 37% reported in the ALASCCA trial. The study emphasizes the importance of upfront genomic testing to facilitate timely treatment decisions and the use of a safe, inexpensive repurposed drug.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Early Detection

MSI/MMR

VERIFIED

EARLY DETECTION

PUBLISHED AUG 21, 2026

⚡ 1

Comparison of PCR-based high-resolution melting curve analysis and immunohistochemistry in detecting microsatellite instability in colorectal cancer: a retrospective consistency study

PCR-based high-resolution melting curve analysis (PCR-HRM) demonstrates high precision and efficiency for detecting microsatellite instability (MSI) in colorectal cancer, simplifying workflow by eliminating post-processing steps required by traditional methods. This study suggests that PCR-HRM may serve as a valuable tool for guiding treatment decisions in colorectal cancer, potentially addressing limitations associated with immunohistochemistry results.

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Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA

Dynamic Circulating Tumor DNA Methylation Monitoring Guiding Postoperative Surveillance in Nonmetastatic Colorectal Cancer: A Prospective, Randomized, Phase III FIND Trial

Dynamic monitoring of circulating tumor DNA (ctDNA) methylation was assessed in a prospective, randomized, Phase III FIND trial for postoperative surveillance in nonmetastatic colorectal cancer. The study indicates that ctDNA methylation can guide surveillance strategies, potentially improving early detection of recurrence.

MSI/MMR

EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C-mutated colorectal cancer (mCRC): Results from phase 3 CodeBreak 300.

In the phase 3 CodeBreak 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C-mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS

Rapid On-Site Next-Generation Sequencing: An Alternative to Single-Gene and Send-Out Testing in Non-Small Cell Lung Cancer and Colorectal Cancer in a Community Pathology Laboratory Setting

Next-generation sequencing (NGS) using the ThermoFisher Oncomine Precision Assay (OPA) was assessed as an alternative to single-gene panels and send-out NGS for identifying actionable mutations in colorectal cancer. The study focused on turnaround time, quantity not sufficient rates, and detection of NCCN-recommended alterations, indicating that OPA may enhance efficiency in mutation identification for targeted therapies in a community pathology laboratory setting.

EGFR

Second-line raltitrexed plus irinotecan (SALIRI) with bevacizumab, cetuximab, or alone for metastatic colorectal cancer: a prospective multicenter cohort-based secondary analysis of the SALLY trial

SALIRI, a combination of raltitrexed and irinotecan, demonstrated feasibility and tolerability as a second-line therapy for metastatic colorectal cancer when combined with bevacizumab or cetuximab in a nonrandomized secondary analysis of the SALLY trial. Favorable outcomes were noted in the SALIRI + cetuximab subgroup, though results should be interpreted with caution due to nonrandom treatment allocation. Further prospective, biomarker-driven randomized studies are needed to clarify the optimal use of SALIRI-based combinations.

RAS

The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage, suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

Methylation

Lactate as a Master Regulator of Immune Suppression: From Metabolic Waste to Epigenetic Checkpoint in Colorectal Cancer

Lactate, a byproduct of aerobic glycolysis driven by KRAS and BRAF mutations in microsatellite-stable colorectal cancer, acts as a central immunosuppressive regulator in the tumor microenvironment. Its accumulation leads to immune suppression, contributing to resistance against immune checkpoint inhibitors. This highlights lactate's role beyond metabolic waste, suggesting potential therapeutic targets to enhance immune response in this cancer subtype.

TMB

Co-Mutation of CREBBP/EP300 and POLE/POLD1 Identifies a TMB-High Subset of MSS CRC

Co-mutation of CREBBP/EP300 and POLE/POLD1 identifies a subset of TMB-high microsatellite stable (MSS) colorectal cancer (CRC). Analysis of multiple public cohorts demonstrated that CREBBP/EP300 mutations are independently associated with TMB-high status ($p < 0.00001$). This finding suggests potential for these mutations as biomarkers for immunotherapy eligibility in MSS CRC.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right

now are **ctDNA**, **MSI/MMR**, **KRAS**, **TP53**, and **BRAF**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-09-13 07:30:02

This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

Colon & Colorectal Cancer — Biomarker Testing Intelligence

About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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https://petroai1492.github.io/delta/timelines/coloncancer_biomarker_timeline.pdf